

1 Pandas

```
[ ]: # Load the weather data.
import pandas as pd

df = pd.read_csv("weather.csv")
print(df)
type(df)
```

```
[2]: # Create a DataFrame from the weather data.
weather_data = [
    ["1/1/2017", 32, 6, 20, "Rain"],
    ["1/2/2017", 35, 7, 30, "Sunny"],
    ["1/3/2017", 28, 2, 45, "Snow"],
    ["1/4/2017", 24, 7, 40, "Snow"],
    ["1/5/2017", 32, 4, 35, "Rain"],
    ["1/6/2017", 31, 2, 20, "Sunny"],
]

df1 = pd.DataFrame(weather_data, columns=["day", "temperature", "windspeed", "humidity", "event"])
df1
```

```
[2]:
```

	day	temperature	windspeed	humidity	event
0	1/1/2017	32	6	20	Rain
1	1/2/2017	35	7	30	Sunny
2	1/3/2017	28	2	45	Snow
3	1/4/2017	24	7	40	Snow
4	1/5/2017	32	4	35	Rain
5	1/6/2017	31	2	20	Sunny

1.1 Working in Rows and Columns

```
[3]: print(df.shape) # to get the shape of the table
```

```
(6, 5)
```

```
[4]: print(df.columns) # to get all the columns of the table
```

```
Index(['day', 'temperature', 'windspeed', 'humidity', 'event'], dtype='str')
```

```
[5]: print(df.head()) # to get the first 5 rows of the table
```

	day	temperature	windspeed	humidity	event
0	1/1/2017	32	6	20	Rain
1	1/2/2017	35	7	30	Sunny
2	1/3/2017	28	2	45	Snow
3	1/4/2017	24	7	40	Snow
4	1/5/2017	32	4	35	Rain

```
[6]: print(df.head(3)) # to get the first 3 rows of the table
```

	day	temperature	windspeed	humidity	event
0	1/1/2017	32	6	20	Rain

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1	1/2/2017	35	7	30	Sunny
2	1/3/2017	28	2	45	Snow

```
[7]: print(df.tail()) # to get the last 5 rows of the table
```

	day	temperature	windspeed	humidity	event
1	1/2/2017	35	7	30	Sunny
2	1/3/2017	28	2	45	Snow
3	1/4/2017	24	7	40	Snow
4	1/5/2017	32	4	35	Rain
5	1/6/2017	31	2	20	Sunny

```
[8]: print(df.tail(2)) # to get the last 2 rows of the table
```

	day	temperature	windspeed	humidity	event
4	1/5/2017	32	4	35	Rain
5	1/6/2017	31	2	20	Sunny

```
[9]: df.sample(3) # to get any random 3 sample
```

```
[9]:
```

	day	temperature	windspeed	humidity	event
0	1/1/2017	32	6	20	Rain
4	1/5/2017	32	4	35	Rain
5	1/6/2017	31	2	20	Sunny

```
[10]: # Select one column.
```

```
print(df.day)
# print(df["day"])
# print(df.loc[:, "day"])
# print(df.iloc[:, 0])
```

0	1/1/2017
1	1/2/2017
2	1/3/2017
3	1/4/2017
4	1/5/2017
5	1/6/2017

Name: day, dtype: str

```
[11]: # Select multiple columns.
```

```
print(df[["day", "temperature"]])
# print(df.loc[:, ["day", "temperature"]])
# print(df.iloc[:, [0, 1]])
```

	day	temperature
0	1/1/2017	32
1	1/2/2017	35
2	1/3/2017	28
3	1/4/2017	24
4	1/5/2017	32
5	1/6/2017	31

```
[12]: df.temperature # Access the temperature column.
```

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```
[12]: 0    32
      1    35
      2    28
      3    24
      4    32
      5    31
      Name: temperature, dtype: int64
```

```
[13]: df.temperature.describe() # to get the description of the row
```

```
[13]: count      6.000000
      mean      30.333333
      std       3.829708
      min      24.000000
      25%      28.750000
      50%      31.500000
      75%      32.000000
      max      35.000000
      Name: temperature, dtype: float64
```

```
[14]: df.temperature.info() # to get the info of a column
```

```
<class 'pandas.Series'>
RangeIndex: 6 entries, 0 to 5
Series name: temperature
Non-Null Count  Dtype
-----
6 non-null      int64
dtypes: int64(1)
memory usage: 180.0 bytes
```

```
[15]: df.iloc[2, 4] # Access by row and column position.
```

```
[15]: 'Snow'
```

```
[16]: df.loc[2, "event"] # Access by row and column label.
```

```
[16]: 'Snow'
```

1.2 Working with operations

```
[17]: df.temperature # Access the temperature column.
```

```
[17]: 0    32
      1    35
      2    28
      3    24
      4    32
      5    31
      Name: temperature, dtype: int64
```

```
[18]: df.temperature.max() # to get the maximum temperature
```

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```
[18]: np.int64(35)
```

```
[19]: df.temperature.min() # to get the minimum temperature
```

```
[19]: np.int64(24)
```

```
[20]: df.temperature.std() # to get the standard deviation temperature
```

```
[20]: np.float64(3.8297084310253524)
```

```
[21]: df.temperature.describe() # Get temperature statistics.
```

```
[21]: count      6.000000
      mean      30.333333
      std       3.829708
      min      24.000000
      25%      28.750000
      50%      31.500000
      75%      32.000000
      max      35.000000
      Name: temperature, dtype: float64
```

```
[22]: df.temperature == df.temperature.max() # Check which row has the maximum temperature.
```

```
[22]: 0    False
      1     True
      2    False
      3    False
      4    False
      5    False
      Name: temperature, dtype: bool
```

```
[23]: df.temperature == df.temperature.min() # Check which row has the minimum temperature.
```

```
[23]: 0    False
      1    False
      2    False
      3     True
      4    False
      5    False
      Name: temperature, dtype: bool
```

```
[24]: df[df.temperature == df.temperature.max()] # to get the maximum temperature row
```

```
[24]:   day  temperature  windspeed  humidity  event
      1  1/2/2017      35          7        30  Sunny
```

```
[25]: df[df.temperature == df.temperature.max()].day # to get the day of maximum temperature
```

```
[25]: 1    1/2/2017
      Name: day, dtype: str
```

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```
[26]: df[df.temperature == df.temperature.max()].windspeed # to get the windspeed of ↵
      ↪maximum temperature
```

```
[26]: 1      7
      Name: windspeed, dtype: int64
```

```
[27]: df.index # to get the index
```

```
[27]: RangeIndex(start=0, stop=6, step=1)
```

```
[28]: df.set_index("day") # to set the day column as index
```

```
[28]:
```

	temperature	windspeed	humidity	event
day				
1/1/2017	32	6	20	Rain
1/2/2017	35	7	30	Sunny
1/3/2017	28	2	45	Snow
1/4/2017	24	7	40	Snow
1/5/2017	32	4	35	Rain
1/6/2017	31	2	20	Sunny

```
[29]: df.set_index("day", inplace=True) # to set the day column as index and modify
      df
```

```
[29]:
```

	temperature	windspeed	humidity	event
day				
1/1/2017	32	6	20	Rain
1/2/2017	35	7	30	Sunny
1/3/2017	28	2	45	Snow
1/4/2017	24	7	40	Snow
1/5/2017	32	4	35	Rain
1/6/2017	31	2	20	Sunny

```
[30]: df.reset_index() # reset to normal indexing
```

```
[30]:
```

	day	temperature	windspeed	humidity	event
0	1/1/2017	32	6	20	Rain
1	1/2/2017	35	7	30	Sunny
2	1/3/2017	28	2	45	Snow
3	1/4/2017	24	7	40	Snow
4	1/5/2017	32	4	35	Rain
5	1/6/2017	31	2	20	Sunny

```
[31]: df.sort_values("humidity") # to sort the data according to humidity in acensending
```

```
[31]:
```

	temperature	windspeed	humidity	event
day				
1/1/2017	32	6	20	Rain
1/6/2017	31	2	20	Sunny
1/2/2017	35	7	30	Sunny
1/5/2017	32	4	35	Rain
1/4/2017	24	7	40	Snow
1/3/2017	28	2	45	Snow

```
[32]: df.sort_values("humidity", ascending=False) # to sort the data according to humidity
      ↪ in descending
```

```
[32]:
```

	temperature	windspeed	humidity	event
day				
1/3/2017	28	2	45	Snow
1/4/2017	24	7	40	Snow
1/5/2017	32	4	35	Rain
1/2/2017	35	7	30	Sunny
1/1/2017	32	6	20	Rain
1/6/2017	31	2	20	Sunny

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